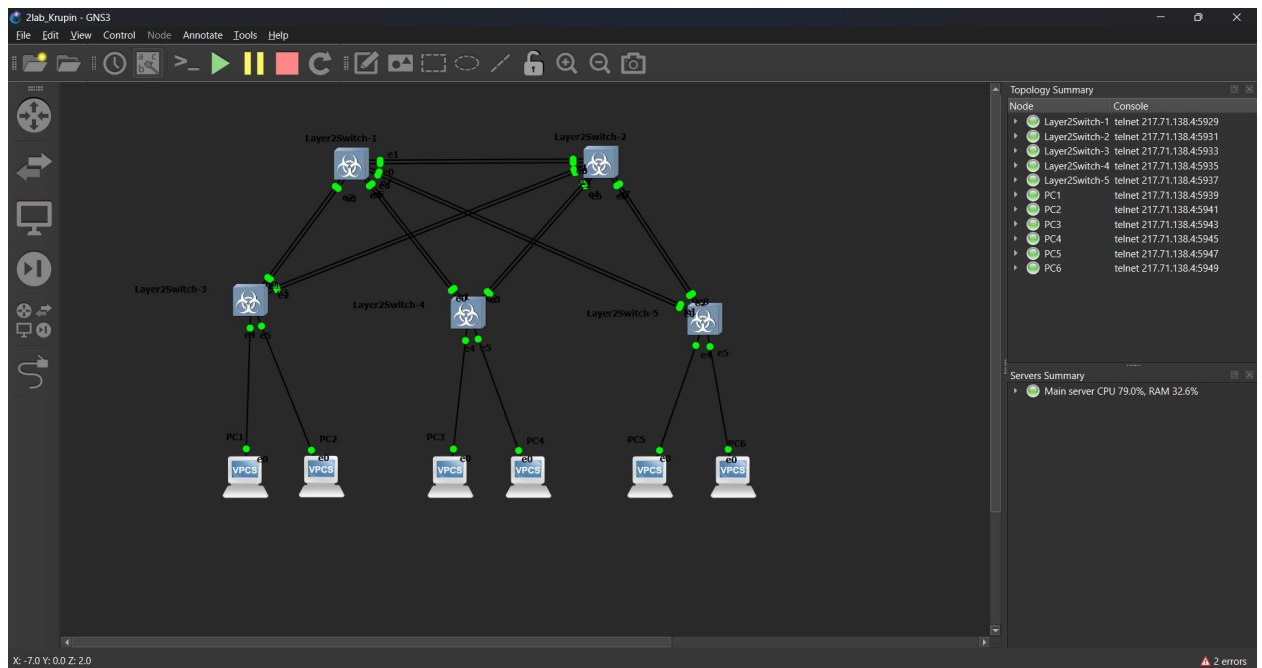




Заходим на Layer2Switch-1

Через консоль

enable

configure terminal

spanning-tree vlan 1 priority 0

end

write memory

show spanning-tree

Назначаем ip PC

ip 192.168.1.1/24

ip 192.168.1.2/24

ip 192.168.1.3/24

ip 192.168.1.4/24

ip 192.168.1.5/24

ip 192.168.1.6/24

Проверяем связь компьютеров с помощью ping

Например на PC1

ping 192.168.1.2

ping 192.168.1.3

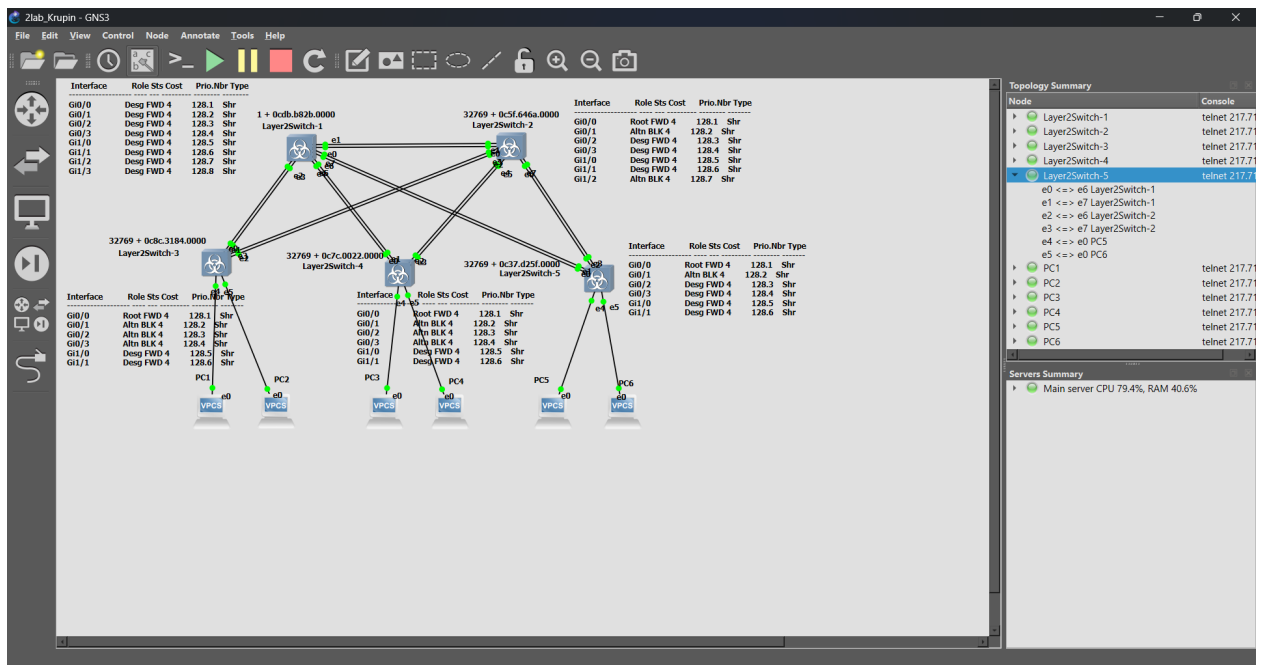
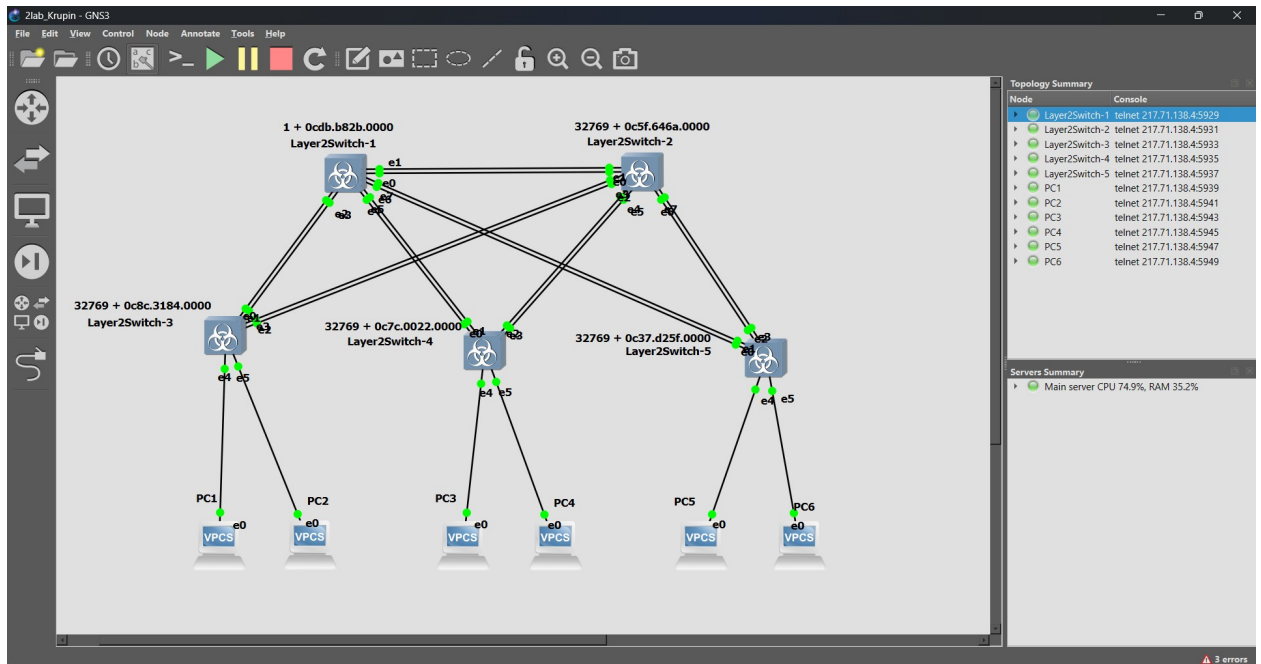
ping 192.168.1.4

ping 192.168.1.5

ping 192.168.1.6

Определить VID и роли портов на каждом коммутаторе

show spanning-tree



Соединение e0:

Wireshark capture showing STP (Spanning Tree Protocol) traffic on interface e0. The capture is titled "Standard input [Layer2Switch-1 Ethernet0 to Layer2Switch-2 Ethernet0]".

The packet list shows multiple STP frames (Type 0) from source 00:5f:64:6a:00:00 to destination 01:00:5e:00:00:00. The packet details pane shows the Spanning Tree Protocol structure:

- Protocol Identifier: Spanning Tree Protocol (0x0000)
- Protocol Version Identifier: Spanning Tree (0)
- BPDU Type: Configuration (0x00)
- BPDU flags: 0x00
- Root Identifier: 0 / 1 / 00:5f:64:6a:00:00
- Root Path Cost: 0
- Bridge Identifier: 0 / 1 / 00:5f:64:6a:00:00
- Port identifier: 0x8001
- Message Age: 0
- Max Age: 20
- Hello Time: 2
- Forward Delay: 15

The packet bytes pane shows the raw data of the frame, including the Ethernet II header and the STP payload.

Соединение e1:

Wireshark capture showing STP (Spanning Tree Protocol) traffic on interface e1. The capture is titled "Standard input [Layer2Switch-1 Ethernet1 to Layer2Switch-2 Ethernet1]".

The packet list shows multiple STP frames (Type 0) from source 00:5f:64:6a:00:00 to destination 01:00:5e:00:00:00. The packet details pane shows the Spanning Tree Protocol structure:

- Protocol Identifier: Spanning Tree Protocol (0x0000)
- Protocol Version Identifier: Spanning Tree (0)
- BPDU Type: Configuration (0x00)
- BPDU flags: 0x00
- Root Identifier: 0 / 1 / 00:5f:64:6a:00:00
- Root Path Cost: 0
- Bridge Identifier: 0 / 1 / 00:5f:64:6a:00:00
- Port identifier: 0x8002
- Message Age: 0
- Max Age: 20
- Hello Time: 2
- Forward Delay: 15

The packet bytes pane shows the raw data of the frame, including the Ethernet II header and the STP payload.

Соединение е2:

Wireshark capture of STP traffic on interface 45UL3.pcapng. The capture shows a series of Spanning Tree Protocol (STP) frames, including BPDUs and configuration messages, transmitted between various network devices (e.g., 00000000000000000000000000000000, 00000000000000000000000000000000).

Key details visible in the packet list and details pane:

- Frame 21: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
- IEEE 802.3 Ethernet
- Logical-link Control
- Spanning Tree Protocol
 - Protocol Identifier: Spanning Tree Protocol (0x0000)
 - Protocol Version Identifier: Spanning Tree (0)
 - BPDU Type: Configuration (0x00)
 - BPDU flags: 0x00
 - Root Identifier: 0 / 1 / 0c:db:b8:2b:00:00
 - Root Path Cost: 0
 - Bridge Identifier: 0 / 1 / 0c:db:b8:2b:00:00
 - Port identifier: 0x8003
 - Message Age: 0
 - Max Age: 20
 - Hello Time: 2
 - Forward Delay: 15

Wireshark Standard input45UL3.pcapng

Соединение е3:

Wireshark capture of STP traffic on interface HQQV3.pcapng. The capture shows a series of Spanning Tree Protocol (STP) frames, including BPDUs and configuration messages, transmitted between various network devices (e.g., 00000000000000000000000000000000, 00000000000000000000000000000000).

Key details visible in the packet list and details pane:

- Frame 27: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
- IEEE 802.3 Ethernet
- Logical-link Control
- Spanning Tree Protocol
 - Protocol Identifier: Spanning Tree Protocol (0x0000)
 - Protocol Version Identifier: Spanning Tree (0)
 - BPDU Type: Configuration (0x00)
 - BPDU flags: 0x00
 - Root Identifier: 0 / 1 / 0c:db:b8:2b:00:00
 - Root Path Cost: 0
 - Bridge Identifier: 0 / 1 / 0c:db:b8:2b:00:00
 - Port identifier: 0x8004
 - Message Age: 0
 - Max Age: 20
 - Hello Time: 2
 - Forward Delay: 15

Wireshark Standard inputHQQV3.pcapng

Соединение е4:

*Standard input [Layer2Switch-1 Ethernet4 to Layer2Switch-4 Ethernet0]

ФайлПравкаВидЗапускЗахватАнализСтатистикаТелефонияБеспроводная связьИнструментыСправка

stp

No.	Time	Delta	Source	Destination	Protocol	Length	Info
12	4.493715	0.001369	0c:db:b8:2b:00:00	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/200/0c:db:b8:2b:00:00 Coa...
13	2.559869	0.150210	0c:7c:00:22:00:00	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/300/0c:7c:00:22:00:00 Coa...
14	2.693838	0.03869	0c:db:b8:2b:00:04	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/300/0c:db:b8:2b:00:00 Coa...
15	2.695120	0.001282	0c:db:b8:2b:00:04	Nearest-Customer-Sr...	STP	60	Conf. Root = 0/1/0c:db:b8:2b:00:00 Coa...
17	4.252338	1.557218	0c:db:b8:2b:00:04	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/100/0c:db:b8:2b:00:00 Coa...
18	4.755244	0.502906	0c:7c:00:22:00:00	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/100/0c:7c:00:22:00:00 Coa...
19	4.756287	0.001143	0c:db:b8:2b:00:04	Nearest-Customer-Sr...	STP	60	Conf. Root = 0/1/0c:db:b8:2b:00:00 Coa...
20	4.768076	0.011689	0c:db:b8:2b:00:04	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/200/0c:db:b8:2b:00:00 Coa...
21	4.781865	0.013789	0c:7c:00:22:00:00	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/200/0c:7c:00:22:00:00 Coa...
22	4.929501	0.147636	0c:7c:00:22:00:00	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/300/0c:7c:00:22:00:00 Coa...
23	5.099451	0.169956	0c:db:b8:2b:00:04	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/300/0c:db:b8:2b:00:00 Coa...
24	5.947637	0.848180	0c:db:b8:2b:00:04	Nearest-Customer-Sr...	STP	60	Conf. Root = 0/1/0c:db:b8:2b:00:00 Coa...
25	6.553139	0.705502	0c:db:b8:2b:00:04	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/100/0c:db:b8:2b:00:00 Coa...
26	6.594876	0.341737	0c:7c:00:22:00:00	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/100/0c:7c:00:22:00:00 Coa...
27	7.015304	0.020488	0c:7c:00:22:00:00	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/200/0c:7c:00:22:00:00 Coa...
28	7.116496	0.121152	0c:db:b8:2b:00:04	Nearest-Customer-Sr...	STP	60	Conf. Root = 0/1/0c:db:b8:2b:00:00 Coa...
30	7.163638	0.013013	0c:7c:00:22:00:00	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/300/0c:7c:00:22:00:00 Coa...

> Frame 28: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

> IEEE 802.3 Ethernet

> Logical-Link Control

> Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

> BPDU flags: 0x00

> Root Identifier: 0 / 1 / 0c:db:b8:2b:00:00

Root Path Cost: 0

> Bridge Identifier: 0 / 1 / 0c:db:b8:2b:00:00

Port identifier: 0x8005

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

0000 01 80 c2 00 00 00 0c db b8 2b 00 04 00 26 42 42 + 48B

0010 02 00 00 00 00 00 00 01 0c db b8 2b 00 00 00 00

0020 00 00 00 01 0c db b8 2b 00 00 80 05 00 00 14 00

0030 02 00 0f 00 00 00 00 00 00 00 00 00

wireshark_Standard input74RT3.pcapng

Пакеты: 30 - Отображено: 28 (93.3%) - Потеряно: 0 (0.0%)

Профиль: Default

Соединение е5:

*Standard input [Layer2Switch-1 Ethernet5 to Layer2Switch-4 Ethernet1]

ФайлПравкаВидЗапускЗахватАнализСтатистикаТелефонияБеспроводная связьИнструментыСправка

stp

No.	Time	Delta	Source	Destination	Protocol	Length	Info
20	5.411793	0.160426	0c:7c:00:22:00:01	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/100/0c:7c:00:22:00:00 Coa...
21	5.497142	0.025046	0c:7c:00:22:00:01	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/200/0c:7c:00:22:00:00 Coa...
22	5.587886	0.150744	0c:7c:00:22:00:01	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/300/0c:7c:00:22:00:00 Coa...
23	5.616399	0.029113	0c:db:b8:2b:00:05	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/100/0c:db:b8:2b:00:00 Coa...
24	5.971158	0.354159	0c:db:b8:2b:00:05	Nearest-Customer-Sr...	STP	60	Conf. Root = 0/1/0c:db:b8:2b:00:00 Coa...
25	6.115623	0.144465	0c:db:b8:2b:00:05	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/200/0c:db:b8:2b:00:00 Coa...
26	6.433365	0.317942	0c:db:b8:2b:00:05	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/300/0c:db:b8:2b:00:00 Coa...
27	7.115155	0.681590	0c:db:b8:2b:00:05	Nearest-Customer-Sr...	STP	60	Conf. Root = 0/1/0c:db:b8:2b:00:00 Coa...
28	7.656033	0.540898	0c:7c:00:22:00:01	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/100/0c:7c:00:22:00:00 Coa...
29	7.694518	0.038485	0c:7c:00:22:00:01	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/200/0c:7c:00:22:00:00 Coa...
30	7.859311	0.164793	0c:7c:00:22:00:01	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/300/0c:7c:00:22:00:00 Coa...
31	7.940380	0.081669	0c:db:b8:2b:00:05	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/100/0c:db:b8:2b:00:00 Coa...
32	8.269784	0.328804	0c:db:b8:2b:00:05	Nearest-Customer-Sr...	STP	60	Conf. Root = 0/1/0c:db:b8:2b:00:00 Coa...
33	8.403779	0.133995	0c:db:b8:2b:00:05	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/200/0c:db:b8:2b:00:00 Coa...
34	8.929132	0.525353	0c:db:b8:2b:00:05	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/100/0c:db:b8:2b:00:00 Coa...
36	9.416223	0.479491	0c:db:b8:2b:00:05	Nearest-Customer-Sr...	STP	60	Conf. Root = 0/1/0c:db:b8:2b:00:00 Coa...
37	9.991021	0.574798	0c:7c:00:22:00:01	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/100/0c:7c:00:22:00:00 Coa...
38	9.997708	0.033187	0c:7c:00:22:00:01	Nearest-Customer-Sr...	STP	60	Conf. Root = 32768/200/0c:7c:00:22:00:00 Coa...

> Frame 36: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

> IEEE 802.3 Ethernet

> Logical-Link Control

> Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

> BPDU flags: 0x00

> Root Identifier: 0 / 1 / 0c:db:b8:2b:00:00

Root Path Cost: 0

> Bridge Identifier: 0 / 1 / 0c:db:b8:2b:00:00

Port identifier: 0x8006

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

0000 01 80 c2 00 00 00 0c db b8 2b 00 05 00 26 42 42 + 48B

0010 03 00 00 00 00 00 00 01 0c db b8 2b 00 00 00 00

0020 00 00 00 01 0c db b8 2b 00 00 80 06 00 00 14 00

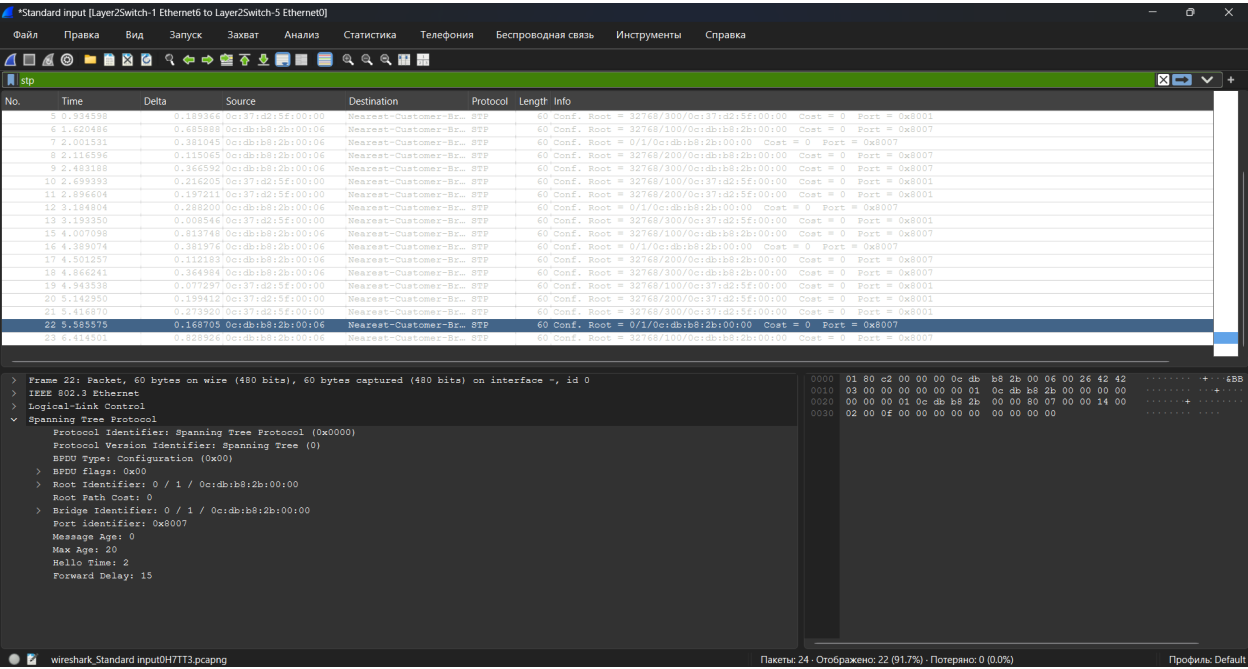
0030 02 00 0f 00 00 00 00 00 00 00 00 00

wireshark_Standard input2HWX73.pcapng

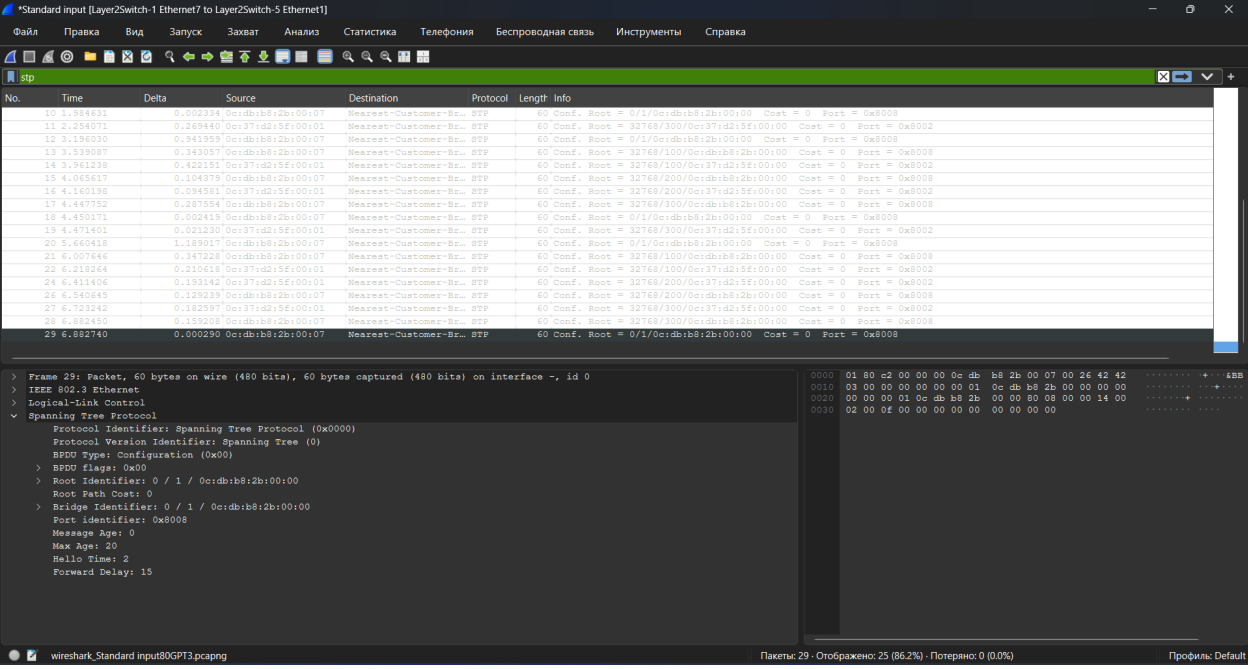
Пакеты: 38 - Отображено: 35 (92.1%) - Потеряно: 0 (0.0%)

Профиль: Default

Соединение еб:



Соединение е7:



Изменить стоимость Root Port

Layer2Switch-4 - PuTTY

```
VLAN0001
Spanning tree enabled protocol ieee
Root ID      Priority      1
            Address      0cdb.b82b.0000
            Cost          4
            Port          1 (GigabitEthernet0/0)
            Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID    Priority      32769 (priority 32768 sys-id-ext 1)
            Address      0c7c.0022.0000
            Hello Time    2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time    300 sec

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi0/0          Root FWD 4          128.1   Shr
Gi0/1          Altn BLK 4          128.2   Shr
Gi0/2          Altn BLK 4          128.3   Shr
Gi0/3          Altn BLK 4          128.4   Shr
Gi1/0          Desg FWD 4          128.5   Shr
Gi1/1          Desg FWD 4          128.6   Shr

--More--
```

enable

configure terminal

interface gigabitEthernet0/0

spanning-tree cost 100

exit

exit

show spanning-tree

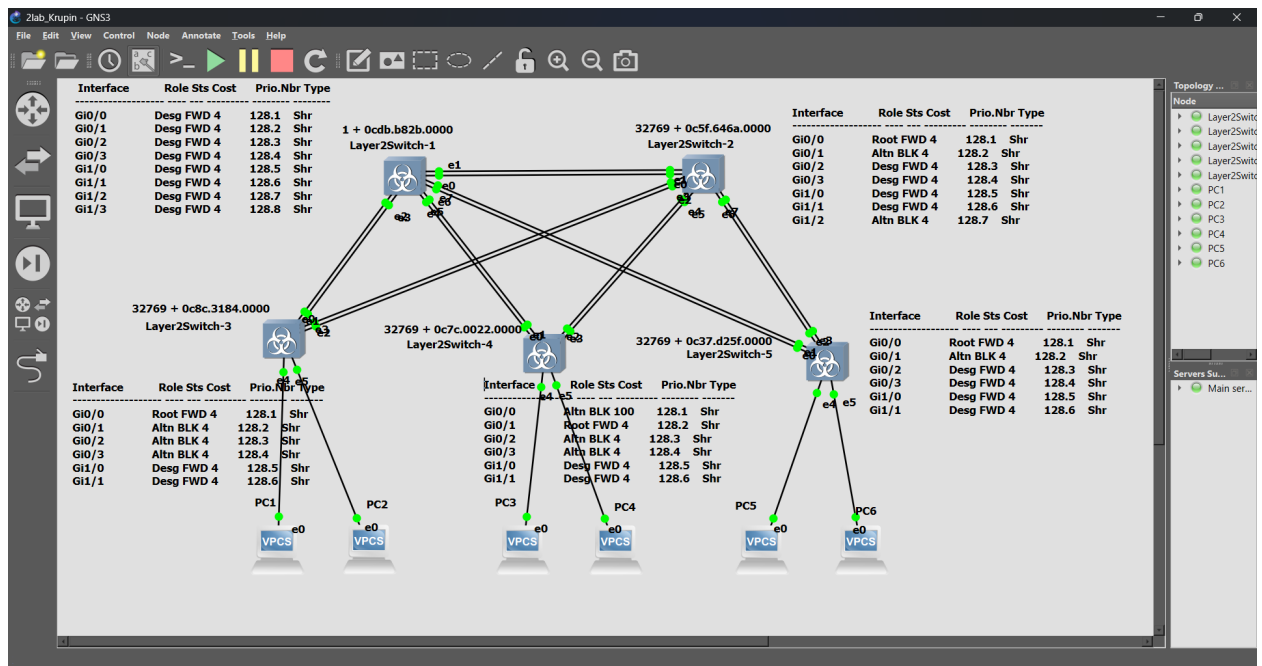

```
Layer2Switch-4 - PuTTY

VLAN0001
Spanning tree enabled protocol ieee
Root ID    Priority    1
           Address    0cdb.b82b.0000
           Cost       4
           Port       2 (GigabitEthernet0/1)
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
           Address    0c7c.0022.0000
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time 300 sec

Interface          Role Sts Cost      Prio.Nbr Type
-----
Gi0/0              Altn BLK 100      128.1   Shr
Gi0/1              Root FWD 4        128.2   Shr
Gi0/2              Altn BLK 4        128.3   Shr
Gi0/3              Altn BLK 4        128.4   Shr
Gi1/0              Desg FWD 4        128.5   Shr
Gi1/1              Desg FWD 4        128.6   Shr

--More--
```



Сохранить конфигурации

На каждом коммутаторе:

enable

show running-config